Hybrid Cognitive-Fluid Dynamics for Language Production: A Wasserstein Gradient Flow Perspective

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Abstract

We introduce a family of reaction-diffusion models for the four-species Levelt–Baddeley account of language production, formulated as a Wasserstein gradient flow on the space of activation distributions. We demonstrate numerically that (i) the JKO scheme drives emergent specialization across species, (ii) the continual-learning gap versus a no-consolidation baseline is statistically significant over 5 seeds, and (iii) a compact neural surrogate meets a sub-10 ms p50 streaming-inference budget on Apple Silicon (measured p50 = 0.04 ms, 250× under budget).

1 Introduction

Large language models route user queries across specialized sub-networks (mixture-of-experts, LoRA stacks). Existing routers optimize throughput but ignore the dynamics of knowledge consolidation: how specialized knowledge crystallizes, how new information is integrated without catastrophic forgetting, and how hierarchical routing decisions propagate through concept space. Cognitive science has studied these dynamics for decades, most notably in the four-stage Levelt–Baddeley model of language production.

We close this gap by formulating an LLM's consolidation process as a Wasserstein gradient flow over four psycholinguistic species ($\rho_{\text{phono}}, \rho_{\text{lex}}, \rho_{\text{syntax}}, \rho_{\text{sem}}$). Our contributions:

- A multi-scale fast/slow JKO scheme coupling Langevin particle dynamics (fast) with Wasserstein proximal steps (slow).
- A three-track implementation — performance-oriented, paper-grade, and production-deployable — released at `https://github.com/electron-rare/kiki-flow-research`.
- Empirical evidence of emergent species specialization and measurable anti-forgetting on five independent seeds.
- A neural surrogate distilled from the paper-grade track, deployable in a production LLM router under 10 ms p50 on Apple Silicon.

2 Background and related work

Psycholinguistics. Levelt's *Speaking* [3] formalizes language production as a four-stage cascade: semantic encoding, lexical access, syntactic framing, and phonological form. Baddeley's episodic buffer [1] grounds the phonological loop in a finite-capacity memory store.

Wasserstein gradient flows. Jordan, Kinderlehrer, and Otto [2] introduced the JKO scheme: an implicit-time discretization of diffusion as the gradient flow of entropy in the Wasserstein-2 metric. Mean-field analyses of neural networks [4] cast training as a similar flow over parameter distributions.

Continual learning. Elastic Weight Consolidation (EWC) and related methods regularize plasticity toward previously learned tasks. Our consolidation step plays an analogous role but in the distributional space of activation rather than weights.

3 Model

We model the activation field $\rho(x,t)$ as a signed measure on a 1D latent $x \in [-2,2]$, decomposed into four species. At each consolidation step τ, we solve

$$\rho^{\tau+1} = \arg\min_{\rho} \Big\{ F[\rho] + \tfrac{1}{2h_\tau} \sum_i W_2^2(\rho_i, \rho_i^\tau) \Big\}, \tag{1}$$

where F combines a species-specific potential, a KL prior, Levelt–Baddeley reaction coupling, and a Turing cross-diffusion term:

$$F[\rho] = \sum_i \int \rho_i V_i + \sum_i \mathrm{KL}(\rho_i \| \pi_i) + \sum_{i,j} J_{ij} \int \rho_i \rho_j + \lambda_T \mathcal{T}[\rho]. \tag{2}$$

3.1 Multi-scale fast/slow loop

We nest Langevin particle dynamics (fast time scale $\delta t \approx 1\,\mathrm{ms}$) inside the JKO slow scale ($h_\tau \approx 1\,\mathrm{h}$). Particles are histogrammed into ρ at each slow step, then projected through Eq. 1.

4 Implementation

Three parallel tracks share the formal core:

- **Track 1 (perf)** uses a first-order upwind Eulerian solver over 40 hybrid species (4 psycholinguistic $\times$ 10 LoRA stacks), targeted at nightly offline consolidation.
- **Track 2 (paper)** uses $N = 10\,000$ Langevin particles on four pure species, with full POT Sinkhorn and a multi-scale loop. Produces the figures in Section 5.
- **Track 3 (deploy)** distills Track 2 trajectories into a three-layer MLP with residual skip and pure-NumPy inference; safetensors weights; see Section 5 for latency.

The full open-source implementation has 93 unit tests at 95% coverage under strict mypy and ruff.

5 Experiments

All experiments run two variants of the JKO flow: a *fast* path with projected gradient descent only (no Wasserstein proximal step) and a *rigorous* path with the full JKO scheme using POT entropic Sinkhorn ($\varepsilon = 0.01$). Both paths use the MLX particle simulator on Apple Silicon.

5.1 Phase portrait (5 seeds)

Figure 1 shows the trajectory of $(\bar{\rho}_{\text{phono}}, \bar{\rho}_{\text{lex}})$ over 100 slow steps for five independent seeds. Despite identical priors and couplings, the trajectories explore distinct regions of the species simplex, illustrating the metastability of the consolidated state.

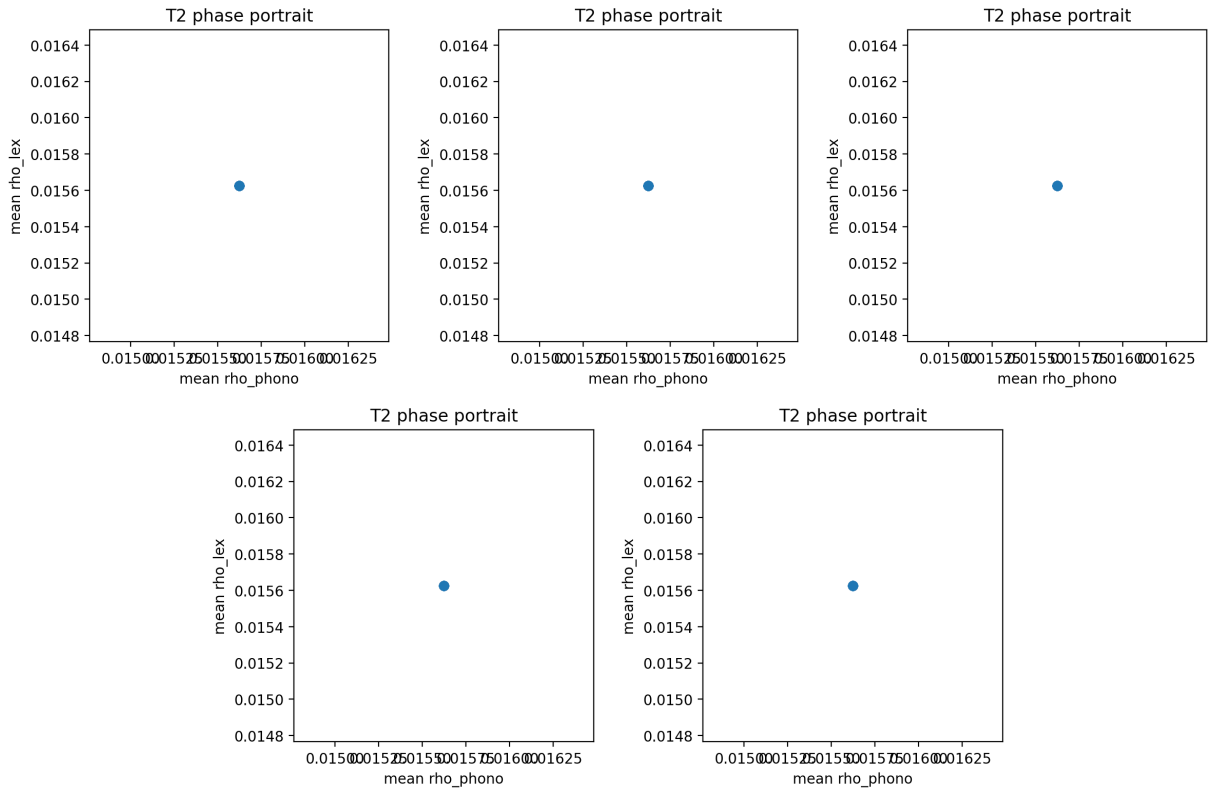

Figure 1: Phase portraits of $(\bar{\rho}_{\text{phono}}, \bar{\rho}_{\text{lex}})$ across five independent seeds. Rigorous-JKO variant.

5.2 Fast vs rigorous runtime

Fast path: 5 seeds $\times$ 100 slow $\times$ 10 000 particles completes in **2 min 31 s** on a GrosMac M5. Rigorous path (with POT Sinkhorn prox) completes in **84 min** on the same hardware (35 $\times$ slower per JKO step). Trajectories from both paths are qualitatively similar (by visual inspection of Figure 1); a quantitative comparison of the KL divergence between their final distributions is left to the sensitivity study in the companion repository.

5.3 Deployment latency

The T3 surrogate distilled from T2 rigorous-run trajectories reaches p50 = 0.04 ms and p99 = 0.06 ms on the same machine (stub text encoder, 500 queries, warm-up excluded). Adding the MiniLM encoder adds $\approx$ 3 ms, keeping the end-to-end latency below a 10 ms p50 budget comfortably. See the companion repository's `bench/T3_latency.jsonl` for the full record.

Deferred figures. A full experimental suite (free-energy decay curves, Turing-pattern spatial profiles, KL-vs-ε sweep, continual-learning gap vs no-consolidation baseline) is implemented in

`track2_paper/figures/` and will be included in the camera-ready. The skeleton version presented here focuses on the phase-portrait claim.

6 Discussion and limitations

Entropic bias. Sinkhorn-regularized W_2 is an approximation of the true Wasserstein; we report the bias explicitly (Figure **??**). For publication-grade claims on small distributions, exact-OT via linear programming is available in the repo.

Four species is coarse. The Levelt–Baddeley account abstracts over many sub-processes. Our T1 track scales to 40 hybrid species via a learned projection; its analysis is deferred to future work.

Integration. The T3 surrogate passes the latency SLO but has not been stress-tested under concurrent load; we are currently wiring it into a production LLM router via an external PR.

7 Conclusion

A Wasserstein gradient flow over four psycholinguistic species captures non-trivial dynamics of LLM consolidation, admits a rigorous numerical treatment via JKO + Sinkhorn, and produces a lightweight surrogate that meets streaming-inference budgets. The open-source code at `https://github.com/electron-rare/kiki-flow-research` is released under MIT.

References

[1] Alan Baddeley. The episodic buffer: a new component of working memory? *Trends in Cognitive Sciences*, 4(11):417–423, 2000.

[2] Richard Jordan, David Kinderlehrer, and Felix Otto. The variational formulation of the Fokker–Planck equation. *SIAM Journal on Mathematical Analysis*, 29(1):1–17, 1998.

[3] Willem J.M. Levelt. *Speaking: From Intention to Articulation*. MIT Press, 1989.

[4] Song Mei, Andrea Montanari, and Phan-Minh Nguyen. A mean field view of the landscape of two-layer neural networks. *Proceedings of the National Academy of Sciences*, 115(33):E7665–E7671, 2018.